ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Electricity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** charge	I = Q/t.
2	**(d)** its length	R is proportional to length.
3	**(b)** 10 ohm	Add.
4	**(d)** Rubber	No free electrons.
5	**(c)** less resistance	Greater area.
6	**(a)** 2 ohm	Half.
7	**(b)** ammeter	A for ampere.
8	**(d)** more resistance	Proportional to length.
9	**(c)** stop glowing	The circuit is broken.
10	**(d)** voltmeter	V for volt.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rate of flow of electric charge (2).	2
12	Opposition offered to the flow of current (1); R = V / I (1).	1+1
13	Rs = R1 + R2 + R3 (2).	2
14	Coulomb (1); About 6.25×10^{18} electrons (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Q = I t (1); Q = 2 x 5 (1); Q = 10 C (1).	1+1+1
16	Length (1); Area of cross-section (1); Nature of the material (1).	1+1+1
17	1 kW = 1000 W (1); 1 h = 3600 s (1); 1 kWh = 3.6×10^6 J (1).	1+1+1
	OR: V = I R (1); V = 0.2 x 10 (1); V = 2 V (1).	1+1+1
18	Rs = R1 + R2 + R3 (1); 3 + 4 + 5 (1); 12 ohm (1).	1+1+1
	OR: Each appliance gets full voltage (1); Each can be switched on or off independently (1); Failure of one does not affect others (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Current is proportional to the potential difference at constant temperature (1); V = I R (1); Graph is a straight line through the origin (1); Slope = V / I (1); Slope gives the resistance (1).	1+1+1+1+1
	OR: E = P t = 0.1 kW x 10 h (1); E = 1 kWh (1); I = P / V (1); I = 100 / 220 (1); I = 0.45 A (1).	1+1+1+1+1
20	Length: R increases with length (1); Area: R decreases with area (1); Nature of the material (1); Temperature (1); R = rho L / A (1).	1+1+1+1+1
	OR: Power is work done per unit time (1); W = V Q = V I t (1); P = V I (1); Substituting V = I R: P = I^2 R (1); Substituting I = V / R: P = V^2 / R (1).	1+1+1+1+1

Mistakes to avoid:

- Adding the reciprocals incorrectly: for resistors in parallel $\frac{1}{R_p} = \sum \frac{1}{R}$ and the final answer is R_p, not the sum of reciprocals.
- Connecting an ammeter in parallel or a voltmeter in series; the ammeter goes in series and the voltmeter in parallel.
- Thinking that resistance is directly proportional to cross-sectional area; $R \propto \frac{1}{A}$ and $R \propto l$.
- Using $H = I^2Rt$ with t in minutes or hours; time must be in seconds (and energy in kWh needs power in kW and time in hours).
- Writing that in a series circuit the voltage across each resistor is the same; the **current** is the same in series and the **voltage** is the same in parallel.
- Forgetting that 1 kWh equals 3.6×10^6 J; and confusing the unit of resistivity (Ω m) with that of resistance (Ω).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Electricity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** increases with temperature	Metals.
2	**(c)** continue to glow	Separate paths.
3	**(a)** in parallel	Across the component.
4	**(b)** ohm metre	Ohm m.
5	**(d)** rate of consumption of energy	P = W/t.
6	**(b)** in series	In the path of current.
7	**(a)** Silver	Lowest resistivity.
8	**(b)** watt	W.
9	**(d)** volt	V.
10	**(c)** very high	10^{12} to 10^{17} ohm m.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Resistance when 1 A flows at a potential difference of 1 V (2).	2
12	1/Rp = 1/R1 + 1/R2 + 1/R3 (2).	2
13	Work done per unit charge in moving between two points (2).	2
14	Ohm metre (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	I = V / R (1); I = 12 / 4 (1); I = 3 A (1).	1+1+1
16	1/R = 1/6 + 1/3 (1); 1/R = 1/2 (1); R = 2 ohm (1).	1+1+1
17	It is connected in series (1); It should not change the current (1); So that it does not reduce the current in the circuit (1).	1+1+1
	OR: R = V / I (1); R = 6 / 3 (1); R = 2 ohm (1).	1+1+1
18	Energy = P x t (1); 0.04 kW x 5 h (1); 0.2 kWh (1).	1+1+1
	OR: High melting point (1); High resistivity (1); Does not oxidise at the working temperature in inert gas (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	R = 2 + 4 + 6 = 12 ohm (1); I = 12 / 12 = 1 A (1); V1 = 2 V (1); V2 = 4 V (1); V3 = 6 V (1).	1+1+1+1+1
	OR: Daily energy = 2 x 3 = 6 kWh (1); 30 days: 180 kWh (1); Cost = 180 x 5 (1); Rs 900 (1); 180 units are used (1).	1+1+1+1+1
20	1/R = 1/10 + 1/15 = 5/30 (1); R = 6 ohm (1); I1 = 6 / 10 = 0.6 A (1); I2 = 6 / 15 = 0.4 A (1); Total current 1 A (1).	1+1+1+1+1
	OR: Each part is half the length (1); Each part has 4 ohm (1); 1/R = 1/4 + 1/4 (1); R = 2 ohm (1); Parallel resistance is less than each part (1).	1+1+1+1+1

Mistakes to avoid:

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- Connecting an ammeter in parallel or a voltmeter in series; the ammeter goes in series and the voltmeter in parallel.
- Thinking that resistance is directly proportional to cross-sectional area; $R \propto \frac{1}{A}$ and $R \propto l$.
- Using $H = I^2Rt$ with t in minutes or hours; time must be in seconds (and energy in kWh needs power in kW and time in hours).
- Writing that in a series circuit the voltage across each resistor is the same; the **current** is the same in series and the **voltage** is the same in parallel.
- Forgetting that 1 kWh equals 3.6×10^6 J; and confusing the unit of resistivity (Ω m) with that of resistance (Ω).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Electricity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** kilowatt hour	kWh.
2	**(b)** V = I R	Ohm's law.
3	**(a)** nickel, chromium and iron	Used in heaters.
4	**(a)** low melting point	It melts quickly.
5	**(c)** a zig-zag line	Standard symbol.
6	**(a)** 6 ohm	Proportional.
7	**(c)** P = V I	Power = V x I.
8	**(d)** current flows	The path is complete.
9	**(d)** ohm	Omega.
10	**(a)** an electric heater	Heater works on I^2 R t.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rate of doing electrical work (1); Watt (1).	1+1
12	V = I R (1); Where R is constant at constant temperature (1).	1+1
13	Nichrome (1); Constantan or manganin (1).	1+1
14	A safety device with a low-melting wire (1); Breaks the circuit on overloading (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	W = Q V (1); W = 5 x 6 (1); W = 30 J (1).	1+1+1
16	P = 1 kW (1); E = 1 x 2 (1); E = 2 kWh (1).	1+1+1
17	In series with the appliance (1); Excess current melts it (1); Breaks the circuit and saves the appliance (1).	1+1+1
	OR: Cell: long and short parallel lines (1); Bulb: a circle with a cross (1); Resistor: a zig-zag line or rectangle (1).	1+1+1
18	P = V I (1); P = 220 x 2 (1); P = 440 W (1).	1+1+1
	OR: H = I^2 R t (1); 4 x 5 x 10 (1); H = 200 J (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Current: same vs divided (1); Voltage: divided vs same (1); Rs greater than each; Rp less than each (1); Failure: breaks all vs others work (1); Use: parallel in homes (1).	1+1+1+1+1
	OR: Copper has low resistivity (1); Less heat is produced in wires (1); Nichrome has high resistivity (1); It can withstand high temperatures (1); It does not oxidise readily (1).	1+1+1+1+1
20	Electrical energy is converted to heat (1); Electrons collide with atoms (1); H = I^2 R t (1); Use: heater (1); Use: electric iron or bulb (1).	1+1+1+1+1
	OR: Cell, key, resistor and ammeter in series (1); Voltmeter is connected across the resistor (1); Positive terminals to the higher potential side (1); Ammeter has low resistance (1); Voltmeter has high resistance (1).	1+1+1+1+1

Mistakes to avoid:

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- Forgetting that 1 kWh equals 3.6×10^6 J; and confusing the unit of resistivity (Ω m) with that of resistance (Ω).

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